Selection Beats Compression: Memory Decision Policies for LLM Agents Under a Bounded Store

YASH RAMCHANDANI
Dept. of Computer Science and Engineering
SVKM's NMIMS, Indore Campus
Indore, India

SIMAR SINGH KHANUJA
Dept. of Computer Science and Engineering
SVKM's NMIMS, Indore Campus
Indore, India

DR. PRAVEEN MODI
Dept. of Computer Science and Engineering
SVKM's NMIMS, Indore Campus
Indore, India

***Abstract*—A long-running LLM agent cannot keep every turn of a conversation in its context window, so a memory layer has to decide what to keep, what to compress, what to forget and what to recall. Tiered memory systems that make these decisions are already in use, but the decision policy itself has rarely been isolated and measured. This paper presents MemGate, an evaluation harness that places competing memory policies on a common accuracy-per-token frontier. Each comparison is set up so that only one factor differs between the two arms. We first show that the accounting normally used in this area, which bounds the context assembled per query but leaves the store unbounded, charges a store-everything baseline nothing for retention and therefore settles the comparison in advance. When the store is also bounded, choosing which turns to forget is worth $+9.37$ answer-recall points over forgetting oldest-first at identical storage, fidelity and retrieval (95% CI $[+5.45, +14.90]$ clustered by conversation, McNemar $p = 2.8\times10^{-5}$), while compressing an evicted turn instead of dropping it loses 7.13 points. We also test the rate–distortion prediction that compression must eventually win. Up to a compression ratio of $182\times$ it does not, although the margin in favour of selection narrows. Charging the budget for the embedding index, which is normally left out, reverses the ranking below about 40 KiB, where a policy that stores no index at all performs best. Finally, an LLM-judged salience scorer performs worse than a regular-expression heuristic by 5.87 points, with the confidence interval excluding zero, whereas a logistic scorer over shallow features and a sentence embedding, trained on LoCoMo and on REALTALK conversations between real people, beats the heuristic by $+14.55$ points on held-out LoCoMo conversations (95% CI $[+10.40, +19.70]$). Across store budgets from 2048 to 16384 tokens its LoCoMo gain stays significant, and on REALTALK it becomes significant once the store holds about 40% of a conversation. Rebuilt in the same harness, with their settings tuned on the test corpus in their favour, MemGPT-style and Mem0-style memory both lose to the learned scorer at every budget from 2048 to 16384 tokens. Held out on BEAM chats of up to ten million tokens it beats retrieval on every corpus tested, and when whole one-million-token chats are typed into the application one message at a time it keeps the answer for 51.9% of questions against 19.2% for retrieval. Through a fixed local reader over all 715 answer-recoverable questions, $+6.0$ points of the selection advantage convert into correct answers (95% CI $[+3.3, +9.1]$), which is 69% of the recall gap in that run. We release the policy as a working chat application that runs offline on a laptop, and report three measurement errors that changed our own results, each found by an invariant rather than by inspection.**

***Index Terms*—large language models, agent memory, context management, retrieval-augmented generation, rate–distortion, empirical evaluation**

I. Introduction

An agent that runs for weeks [1], [2] accumulates a history far larger than any context window [3], so something must decide what enters the prompt each turn. A longer window does not remove the problem: accuracy falls 30–50% well before the documented limit across frontier models [4], and position inside a long context changes how much the model uses [5]. These are properties of attention over long inputs, not gaps in training, so a larger window moves the difficulty rather than solving it. The limiting factor is curation, not capacity.

The cost of poor curation can be measured. On LongMemEval, oracle retrieval scores about 92% where the same model retrieving for itself scores about 58% [6]. The model answers correctly once the right evidence is placed in front of it, so this 34-point difference is a selection failure rather than a reasoning failure.

Systems that address selection already exist, and agent architectures now treat memory as a first-class module [7]. Tiered memory with summarisation and forgetting is used in MemGPT [8], MemoryBank [9] and the architectures reviewed in [10]. What has not been established is whether the individual mechanisms inside those systems earn their cost, since they are evaluated as complete systems rather than as separate decisions. A recent survey states the gap directly: current systems "do not solve the fundamental challenge: deciding what to remember and what to forget" [10]. A rate–distortion framing [11], [12] has been proposed for this family of decisions [13], but it provides an analytical framework rather than a measured system.

This paper measures the decision. Our research question is: how much of the memory-selection gap can a better decision policy recover, and under what budget does compression become worthwhile?

Our contributions are as follows. (1) A storage-budgeted protocol correcting an accounting error common in this area: bounding only the assembled context charges a store-everything baseline nothing for retention, settling the comparison before it is run (Section VI-A). (2) Evidence that the gain is selection, not compression — at identical storage, fidelity and retrieval, scored forgetting is worth $+9.37$ answer-recall points over FIFO while compressing an evicted turn loses 7.13

(Sections VI-B and VI-C). (3) A test of the rate–distortion crossover, to $182\times$ compression (Section VI-D). (4) Byte-denominated accounting: the embedding index outweighs the text it indexes by about $12\times$ and reverses the policy ranking below roughly 40 KiB (Section VI-E). (5) A negative result for LLM-judged salience — as a single-component swap, a 0.5B judge loses to a hand-written heuristic by 5.87 points (Section VI-F). (6) A learned scorer trained on two corpora, one LLM-generated and one written by people, with the direction in which it transfers and the direction in which it does not (Section VI-G). (7) An end-task evaluation through a fixed reader, measuring how much of the recall advantage becomes correct answers (Section VI-H). (8) Comparisons with MemGPT-style and Mem0-style memory rebuilt in the same harness (Section VI-I). (9) A scale test on chats of up to ten million tokens, and a scorer retrained on 14.6 times the data, including the finding that more data alone did not help (Sections VI-J and VI-K). (10) A working application built on the same code paths as the experiments, with the scoring rules and equations it runs (Sections III-C and VII). (11) Three measurement errors in our own results, with the checks that caught them (Section VIII).

Five of these eleven results are negative or corrective. We consider this a useful outcome. Described only as a three-tier memory layer, the system would duplicate work that has already been done; measured against controls, it indicates which parts of that design actually pay for themselves.

II. Related Work

MemGPT [8] pages a virtual context against external storage under the model's own control; MemoryBank [9] adds an Ebbinghaus forgetting curve; Generative Agents [14] score memories by recency, importance and relevance; A-MEM [15] organises them as interlinked notes, and Mem0 [16] extracts and updates facts with an LLM. The design echoes the multi-store model of human memory [17]. This work shows tiering and forgetting are practical, but because these systems are evaluated end to end, the contribution of any single decision inside them cannot be separated from the pipeline. Our harness is built to make that separation possible.

A second line compresses the prompt or the attention cache rather than choosing between stored items: LLMLingua [18] prunes tokens by perplexity, Gist tokens [19] and AutoCompressors [20] learn dense summaries, the in-context autoencoder [21] compresses a context into memory slots, and StreamingLLM [22] and H_2O [23] evict from the KV cache by attention-derived scores. Architectural memories — Transformer memories that compress [24] or index [25], [26] past activations — change the model itself. All operate within a single forward pass over a contiguous context, or require training the model. The decision studied here is upstream and persistent: it fixes which items survive across sessions, where an evicted item cannot be recovered from the cache at all.

Retrieval augmentation [27]–[29] is the store-everything baseline these systems are measured against, and the RAG arm of our harness. Choosing what to evict from a bounded store is also the classical page-replacement problem [30], [31], with the difference that the future accesses here are questions no policy can see.

LoCoMo [32] supplies multi-session dialogues annotated with gold evidence turn identifiers, which is what lets a retrieval-level metric be computed without a generative model; LongMemEval [6] isolates several memory abilities and is the source of the gap quoted above, and LongBench [33] measures long-context understanding without a memory layer. REALTALK [34] applies LoCoMo's annotation schema to 21 days of messaging between real people, which makes it the natural test of whether a scorer fitted on LoCoMo learned what matters or only what LLM-generated dialogue looks like. We propose neither a new architecture nor a new benchmark: the contribution is a measurement protocol and its results, including a case where correcting the protocol reverses the conclusion.

III. The MemGate Harness

MemGate is placed between an agent and a frozen LLM, and controls only what enters the prompt (Fig. 1). The base model is never trained or fine-tuned and is treated as a black box, as in retrieval augmentation [27], so any difference we report can be attributed to the memory policy.

A. Tiers as a lifecycle

The store has three tiers, but they are organised as a lifecycle rather than as a partition. Every turn enters Tier 1 verbatim, and the routing decision is taken once, at eviction, when the budget binds. Writing u for the utility assigned to an item by the scorer:

$$\begin{array}{lll} u \geq \tau_{\text{fact}} & \rightarrow & \text{Tier 3: atomic fact + vector} \\ u \geq \tau_{\text{low}} & \rightarrow & \text{Tier 2: compressed gist} \\ \text{otherwise} & \rightarrow & \text{dropped} \end{array} \tag{1}$$

with $\tau_{\text{fact}} = 0.45$ and $\tau_{\text{low}} = 0.15$; an item with $u \geq \tau_{\text{fact}} + 0.2$ also leaves a gist in Tier 2. The tiers follow the working-set [31] and multi-store [17] views of memory. Tier 1 holds recent verbatim turns and answers the question *when*; Tier 2 holds compressed gists and answers *gist*; Tier 3 holds durable facts with embeddings, answers *what*, and enters the context only when retrieval matches. Scoring each item once, at eviction, is what makes the decision point separable, since it gives a single site in the pipeline that a controlled swap can target.

B. Policies

The arms are P0 (sliding window, recency, verbatim), RAG (store-all, FIFO, verbatim), P1 (threshold, scored at ingest), P1-A (adaptive: score orders eviction, compress), P1-S (select: score orders eviction, drop), P2 (0.5B judge), P3 (learned, never fitted on the conversation it scores) and an oracle. Two require comment. RAG is the no-policy control: it stores everything, retrieves the top k items, and evicts FIFO when the store is full. It is not a weak baseline but the arm that a decision policy has to beat in order to have contributed anything. P0 is given the same token budget as MemGate, so

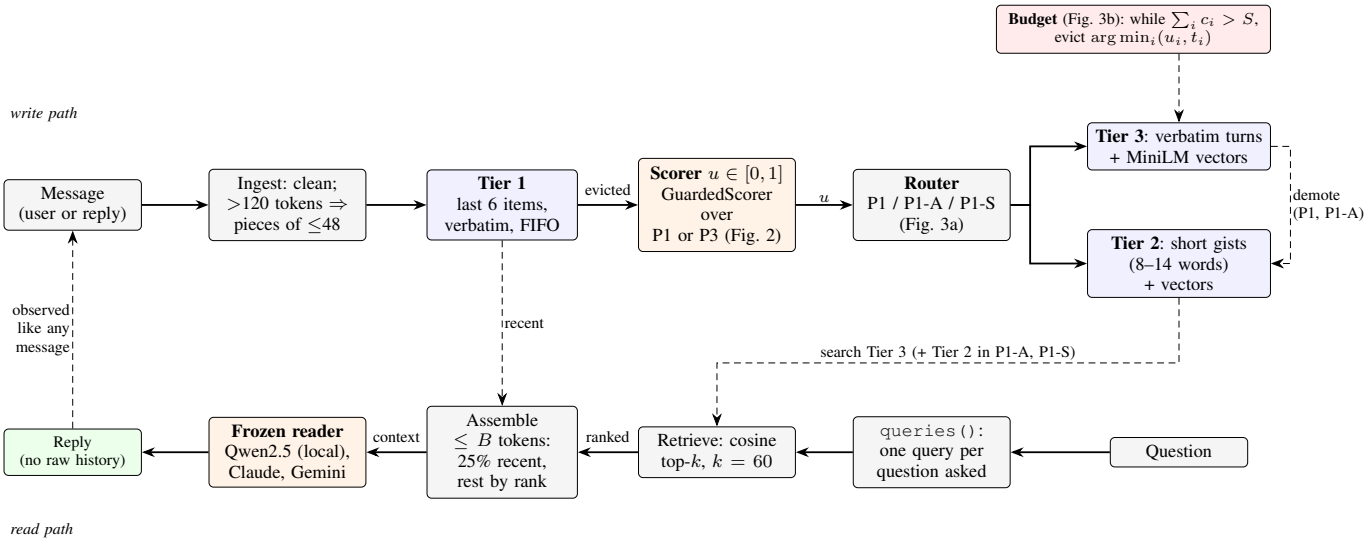


Fig. 1. System architecture. Write path (top): every message, the reader's replies included, enters Tier 1 verbatim; when it leaves Tier 1 it is scored once and routed, and the budget enforcer holds the store at S. (The application adds an entry gate in front of Tier 1, Section VII; the experiments do not use it.) Read path (bottom): each question in a message is its own retrieval query, and the reader receives only the assembled context of at most B tokens, never the raw history. Dashed arrows carry state rather than the item being processed.

the only difference between them is which turns are selected, not how many tokens may be spent.

C. Scoring, routing and the storage budget

Every equation in this subsection is transcribed from the code that produced the results: the modules `scoring`, `pretrained`, `ingest`, `store` and `policies`, and the two training scripts. A message m is always scored with its speaker tag removed, and the scorer sees its text and speaker only: no question, evidence label or future turn.

The P1 heuristic. Let $W(m)$ be the words of m, D a list of 28 durability cues (*deadline, budget, prefer, born, lives, works*, ...), and $h(m) = |W(m) \cap D|$. With indicators δ_{num} (a digit), δ_{tmp} (an amount of money, or a month prefix), δ_{prop} (a capitalised word not opening a sentence), $\delta_{?}$ (ends with a question mark), δ_{asst} (written by the assistant) and δ_{upd} (a correction cue such as *actually* or *no longer* together with a digit or month),

$$\begin{aligned} s(m) &= 0.15 + \min(0.35,\, 0.18\,h) + 0.20\,\delta_{\text{num}} + 0.15\,\delta_{\text{tmp}} \\ &\quad + 0.12\,\delta_{\text{prop}} - 0.12\,\delta_{?} - 0.10\,\delta_{\text{asst}} + 0.10\,\delta_{\text{upd}}, \\ u_{\text{P1}}(m) &= \begin{cases} 0.02, & m \text{ is filler}, \\ \min\big(1, \max(0,\, s(m))\big), & \text{otherwise}, \end{cases} \end{aligned} \tag{2}$$

where a message is filler when it is an acknowledgement or greeting from a closed list, or has at most two words. The weights were set by hand and never fitted.

The learned scorer P3. Its input concatenates 15 hand features $\phi(m)$ (word and character counts, digit, month, money, proper noun, question, final period, $h(m)$, correction cue, filler, assistant speaker, share of words of at most three letters, capital letters, clock time) with the 384-dimensional unit-norm MiniLM embedding $e(m)$ [35], [36]:

$$\begin{aligned} x(m) &= \big[\phi(m);\, e(m)\big] \in \mathbb{R}^{399}, \\ z(m) &= b + \sum_{j=1}^{399} w_j\, \frac{x_j(m) - \mu_j}{\sigma_j}, \quad u_{\text{P3}}(m) = \frac{1}{1 + e^{-z(m)}}, \end{aligned} \tag{3}$$

with μ, σ the training mean and standard deviation of each input. The label is distant supervision [37]: $y_i = 1$ when any question cites turn i as evidence. With N training turns from corpora c grouped into families $f \in F$ (LoCoMo, REALTALK, BEAM, MemBench, everyday chat), the fit is an ℓ_2-regularised, class-balanced, sample-weighted logistic regression fitted with scikit-learn [38],

$$\begin{aligned} &\min_{\mathbf{w}, b}\ \tfrac{1}{2}\|\mathbf{w}\|^2 + C \sum_i \omega_i\, \kappa_{y_i}\, \ell\big(y_i,\, u_{\text{P3}}(m_i)\big), \\ &\omega_i = \frac{N\, \pi_{c(i)}}{|F|\, n_{c(i)}}, \qquad \kappa_y = \frac{N}{2N_y}, \end{aligned} \tag{4}$$

where ℓ is the log loss, $C = 1$, n_c is the number of turns of corpus c, and π_c is the share of its family's weight that corpus c receives: n_c/n_f (by turns, the recipe of the PR #12 model), or P_c/P_f (by evidence turns P, used for the larger model). Every family therefore weighs the same, whatever its size. Generated small-talk lines enter as negatives with total weight $0.05N$.

The shipped ensemble. The shipped scorer averages the logits of two such models, A (the PR #12 recipe) and B (all 856,876 turns). Because each logit is linear in x, the mean is again a single logistic regression, which is how it is stored

(Fig. 4):

$$\bar{z}(m) = \tfrac{1}{2}\big(z_A(m) + z_B(m)\big) = \bar{b} + \bar{\mathbf{w}}^\top x(m),$$
$$\bar{w}_j = \tfrac{1}{2}\sum_{k\in\{A,B\}} \frac{w_j^{(k)}}{\sigma_j^{(k)}}, \quad \bar{b} = \tfrac{1}{2}\sum_k \Big(b^{(k)} - \sum_j \frac{w_j^{(k)}\mu_j^{(k)}}{\sigma_j^{(k)}}\Big). \tag{5}$$

P1-S only orders by u, but the thresholds of (1) and the guards below read u as absolute. The logit is therefore mapped by an affine function that leaves every ranking unchanged, in the manner of Platt scaling [39],

$$z' = \alpha\bar{z} + \beta, \qquad \beta = \operatorname{logit}\tau_{\text{fact}} - \alpha\, q_{\text{fact}},$$
$$\alpha = \frac{\operatorname{logit}\tau_{\text{fact}} - \operatorname{logit}\tau_{\text{low}}}{q_{\text{fact}} - q_{\text{low}}}, \tag{6}$$

where q_{fact} and q_{low} are the quantiles of $\bar{z}$ over LoCoMo and REALTALK turns that put 36.48% of them above τ_{fact} and 71.85% above τ_{low}, the shares of the first shipped model.

The application's guard. The application wraps its scorer in GuardedScorer (Fig. 2). Each guard fires on one kind of input the scorers were not built for (no words, small talk, code, non-Latin script, repetition), and ordinary English passes straight through. One further rule corrects a measured weakness of the learned scorer: trained with BEAM, whose long answers are full of numbers nobody asks about, it gave a short dated fact the score of a greeting. For a message of $n(m) \le 30$ words that names a month, a clock time or an amount,

$$u_{\text{G}}(m) = \max\big(u_{\text{P3}}(m),\, u_{\text{P1}}(m)\big), \tag{7}$$

and $u_{\text{G}} = u_{\text{P3}}$ otherwise. The experiments never use the guard.

The storage budget. Each stored item i has a cost

$$c_i = \begin{cases} \text{tok}(i) & \text{(tokens)}, \\ \text{bytes}(i) + 1536\,\mathbb{1}[i \text{ has a vector}] & \text{(bytes)}, \end{cases} \tag{8}$$

and after every write the store enforces

$$\sum_{i\in T_1\cup T_2\cup T_3} c_i \;\le\; S \tag{9}$$

by evicting, while it is violated,

$$v = \arg\min_{i\in T_3}\,(u_i,\, t_i) \tag{10}$$

in lexicographic order, t_i being the turn index, so the oldest item breaks a tie (Fig. 3b). Superseded records are reclaimed before any eviction. P1 and P1-A demote v to a Tier 2 gist of at most eight words when that gist costs less than v, and drop it otherwise; P1-S always drops it; when Tier 3 is empty the same rule is applied to Tier 2. Tier 1, the six-item live buffer, is never evicted. RAG uses $\arg\min_i t_i$, first in first out, and has no scorer. With equal costs, repeated application of (10) leaves P1-S holding exactly the highest-utility items seen so far, and RAG the newest ones, the FIFO rule of page replacement [30]; this is the only difference between them.

Context assembly. For a question q and context budget B (2048 tokens in every experiment), the most recent Tier 1 items are packed into $\lfloor B/4 \rfloor$ tokens. In P1-A and P1-S the remainder is filled greedily from one pool of Tier 2 and Tier 3 items,

$$R_k(q) = \text{top}_k\,\{\, i : \cos\big(e(q), e(i)\big) > 0 \,\}, \qquad k = 60, \tag{11}$$

ranked by cosine similarity. P1 instead gives Tier 2 a fixed quarter of B, packed by information density, and fills the last half from Tier 3 by (11). When a message asks several questions, each is its own query and the rankings are interleaved, best of each first.

Store maintenance. Three rules keep the store consistent, and none reads an evaluation label. A write to Tier 3 whose embedding has $\cos \ge 0.92$ with an active fact is folded into that fact instead of being stored twice. A write the P1 heuristic labels a correction (Section III-C: a correction cue with a digit or month) retires the single most similar active fact with $\cos \ge 0.60$. When Tier 2 outgrows its share, its oldest half is consolidated: items with $u \ge 0.5$ are promoted to Tier 3 verbatim and the rest are merged into one coarser gist.

D. One variable per comparison

The experimental design lies in the set of contrasts rather than in the set of policies. Each contrast holds everything constant except one factor:

Contrast	Held constant	Varies
P0 vs. RAG	retention, storage	read path
RAG vs. P1-S	storage, fidelity, retrieval	**what is forgotten**
P1-S vs. P1-A	storage, selection	**fidelity**

The second contrast is the one this paper depends on. RAG and P1-S hold the same verbatim turns, in the same embedding space, retrieved by the same procedure, under the same storage cap. The only difference is which turns are discarded when the cap binds: oldest-first for RAG, lowest-utility-first for P1-S. Any gap between the two is therefore attributable to the decision policy.

IV. Metrics

We report two retrieval-level metrics. Evidence recall, also called strict recall, is the fraction of questions for which every gold evidence turn reached the assembled context. Answer recall is the fraction for which the answer text survived into the context, a string-containment test in the spirit of extractive QA scoring [40], computed over the 715 of 1,527 questions whose answer is literally recoverable from the evidence turns.

The distinction is load-bearing. Evidence recall credits an identifier: a compressed gist retains the identifier of a turn whose text it discarded, and records a hit with none of the answer behind it. In development we saw evidence recall rise from 8.1% to 14.8% while answer recall stayed at 12.8% — the gain was entirely identifier bookkeeping. Dropping against compressing is precisely the comparison on which the two diverge, so reporting evidence recall alone would have produced the opposite conclusion. We treat answer recall as primary and report evidence recall alongside it to keep the divergence visible.

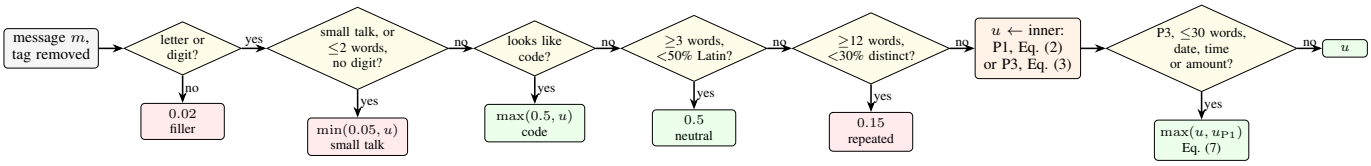


Fig. 2. How the application scores a message (GuardedScorer). Guards are tried left to right; a message that none of them covers is scored by the inner scorer, and a short message naming a date, time or amount is floored at the P1 score, Eq. (7).

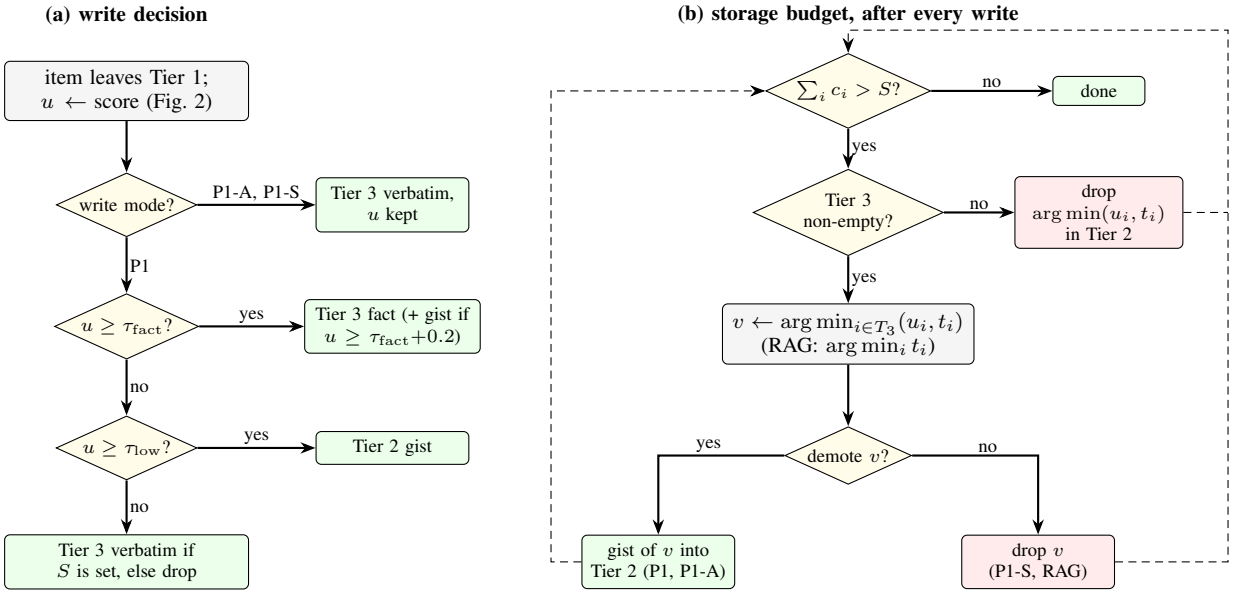


Fig. 3. Where the keep, compress and forget decisions are taken. (a) Routing of an item leaving Tier 1, Eq. (1): threshold mode (P1) decides at ingest; the adaptive modes (P1-A, P1-S) keep everything verbatim and let u order the eviction queue. (b) The budget loop of Eqs. (9)–(10), run after every write. P1-A and P1-S differ only at the "demote" decision, and P1-S and RAG only in the eviction key.

Both are forms of context recall, an upper bound on end-task accuracy rather than accuracy itself. Their advantage here is that they are deterministic and do not confound the memory policy with the reader's reasoning, unlike LLM-judged accuracy [41]; Section VI-H measures accuracy itself.

V. Experimental Setup

We use LoCoMo [32]: 10 multi-session conversations, 5,882 turns, 1,527 scoreable questions. Category-5 (adversarial) items are excluded — they carry evidence, but the correct behaviour is to decline, so retrieving it would not be success.

Retrieval and semantic supersession use all-MiniLM-L6-v2 [36] at 384 dimensions, and budgets are counted with `cl100k_base`, so a budget is an exact token count. The P2 judge is Qwen2.5-0.5B-Instruct [42], run locally; salience is read as a continuous expectation over digit-token logits in one forward pass rather than generated and parsed, which is faster, cannot fail to parse, and leaves no ties in the eviction ranking.

All intervals are paired bootstrap [43] with $B = 10{,}000$ resamples drawn at the conversation level, with an exact McNemar test [44], [45]. The resampling unit is not a detail: 1,527 questions from 10 conversations are not 1,527 independent trials [46], and one of our own claims does not survive the correction (Section VI-C). Every model is a pinned local copy with the hub disabled at import, and all numeric tables are generated from the committed result files rather than transcribed.

VI. Results

A. Storage was never budgeted

Experiments in this area cap the context assembled per query and leave the store unbounded, and our own experiments did the same until this point. Under that accounting a store-everything baseline is charged nothing for holding all 419 turns of a conversation, so forgetting can only lose information. Run under those conditions, our own ablation reported the tiering and compression machinery as a net loss against simply keeping everything and retrieving, and that result was correct given how the cost was being counted.

Imposing a storage budget S changes the question from what fits in the context to what is worth keeping at all. Table I sweeps S with the context budget fixed at 2048 tokens. The ranking under the corrected accounting differs from the ranking under the original one. This is the most consequential observation in the paper, because an evaluation protocol that omits a cost answers a different question from the one it appears to ask.

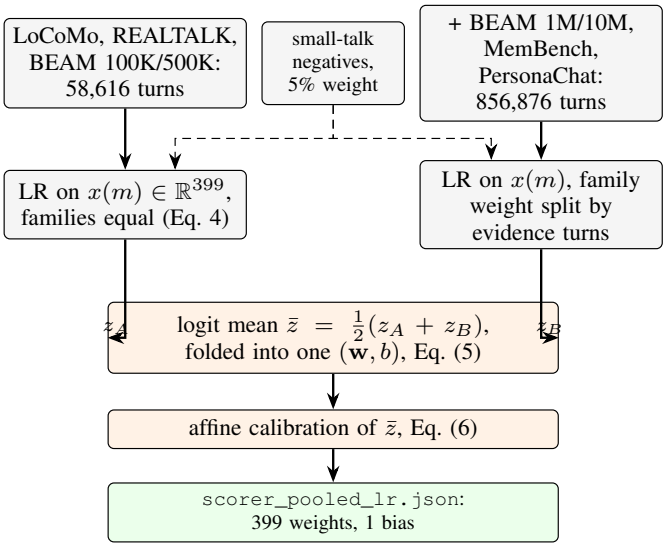


Fig. 4. How the shipped scorer is built: two logistic regressions, Eq. (4), averaged in logit space and folded into one, Eq. (5), then calibrated, Eq. (6). It ships as 399 weights and a bias in plain JSON.

TABLE I
RECALL AGAINST THE STORAGE BUDGET S IN TOKENS, WITH THE CONTEXT BUDGET FIXED AT 2048. Δ IS P1-S MINUS RAG ON ANSWER RECALL: IDENTICAL STORAGE, IDENTICAL FIDELITY, IDENTICAL RETRIEVAL, DIFFERING ONLY IN WHICH TURNS ARE FORGOTTEN.

	Evidence recall (%)			Answer recall (%)			
S	RAG	P1-A	P1-S	RAG	P1-A	P1-S	Δ
1024	4.3	9.3	5.3	9.1	11.9	**15.8**	+6.7
2048	10.1	19.4	11.8	18.3	22.7	**27.8**	+9.5
4096	17.8	31.8	20.2	28.1	30.3	**37.5**	+9.4
8192	30.6	45.6	38.6	44.3	41.0	**57.8**	+13.4
16384	53.3	58.2	57.9	66.4	63.1	**71.3**	+4.9

B. Selection pays; compression does not

Table I reports both metrics across the sweep. P1-S, which drops the lowest-utility item when the cap binds, leads on answer recall at every budget. P1-A, which compresses that item instead of dropping it, leads on evidence recall at every budget, yet at $S = 8192$ trails even RAG on answer recall, by 3.4 points.

This is the identifier problem of Section IV in practice. A demoted gist retains the evidence identifiers of a turn whose text has been discarded. An evaluation reporting only evidence recall would conclude that compression is the better use of a bounded store, and an evaluation that checks whether the answer text survived reaches the opposite conclusion.

C. Statistical significance

Table II reports each contrast with intervals resampled at the conversation level. The answer-recall gain of scored eviction over FIFO is $+9.37$ points with the interval excluding zero. The corresponding evidence-recall gain is $+2.42$ points and does not survive clustering ($p = 0.08$), so we do not claim it. We report it here rather than omitting it, since a claim that fails a correctness check is itself information about the check.

The P1-S against P1-A rows show the divergence with signs attached: $+7.13$ on answer recall and -11.53 on evidence recall, both significant. The two metrics therefore disagree in direction and not only in magnitude.

D. The rate–distortion crossover is not reached

Rate–distortion reasoning predicts compression must eventually beat selection, since a sufficiently compressed record of everything should outperform a faithful record of a subset [13]. Sweeping S downward cannot test this: every policy hits the floor first, near 0.2% evidence recall at $S = 256$. The missing axis is length, so we concatenate up to ten LoCoMo conversations — 186K tokens — holding S fixed, which raises the compression ratio to $182\times$. Table III reports no crossover: the margin of selection over compression narrows but stays positive, and the best operating point stays at the vertex, a subset at full fidelity.

One caveat. A concatenated stream is not a natural long conversation — each question concerns one constituent and the others act as distractors — so this is a stress test of retention rather than a long-dialogue benchmark. We falsify the crossover on this axis to this ratio, and claim nothing beyond.

E. The embedding index is not free

Denominating storage in text tokens leaves out the index that makes retrieval possible. A 384-dimensional `float32` vector occupies 1,536 B, while a LoCoMo turn averages about 130 B of text. The index therefore outweighs the content it indexes by roughly a factor of 12, and the limiting resource becomes the number of stored items rather than their length.

Table IV repeats the sweep with the budget denominated in bytes and each retrievable item charged for its vector. Two results follow. First, the selection finding of Section VI-B is robust to the change of units, since P1-S leads at every budget above 32 KiB. Second, below approximately 40 KiB an embedding index cannot pay for its own storage: P0, which maintains no index, performs best. A protocol that counts only text cannot observe this regime, which is the regime that a memory layer on a constrained device would occupy.

F. Scorer analysis

If the decision is where the value lies, then the scorer that makes it is the component to improve. Table V removes one mechanism at a time. Retrieval dominates, at -31.6 evidence points, and scored eviction is worth 6.4 points. The demotion row is marked because it shows the same divergence in miniature: removing demotion costs 11.5 evidence points and gains 7.1 answer points.

Table VI varies only the scorer, everything else fixed at the configuration of Section VI-B. Three results follow, all significant under clustering. Scoring is worth doing: the hand-written heuristic gains $+11.19$ answer points over unscored recency. The LLM judge, the approach of LLM-as-a-judge

TABLE II

PAIRED BOOTSTRAP ($B = 10{,}000$) RESAMPLED BY CONVERSATION, WITH EXACT MCNEMAR, AT $S = 4096$. THE SECOND ROW IS THE CLAIM WE DECLINE TO MAKE.

Comparison	Metric	Δ (pts)	95% CI (clustered)	McNemar p	
P1-S vs. RAG (FIFO)	answer	$\mathbf{+9.37}$	$[+5.45, +14.90]$	2.8×10^{-5}	✓
P1-S vs. RAG (FIFO)	evidence	$+2.42$	$[-1.48, +6.43]$	0.08	N.S.
P1-S vs. P1-A (compress)	answer	$\mathbf{+7.13}$	$[+4.32, +10.28]$	2.1×10^{-5}	✓
P1-S vs. P1-A (compress)	evidence	$\mathbf{-11.53}$	$[-14.73, -8.47]$	1.8×10^{-23}	✓
P1-S vs. P0 (recency)	answer	$\mathbf{+19.16}$	$[+15.42, +24.13]$	8.2×10^{-19}	✓
P1-S vs. P0 (recency)	evidence	$\mathbf{+10.15}$	$[+6.58, +13.84]$	1.2×10^{-15}	✓

TABLE III

ANSWER RECALL AGAINST COMPRESSION RATIO AT $S = 2048$. THE RIGHTMOST COLUMN IS THE MARGIN OF DROPPING OVER COMPRESSING.

Ratio	RAG	P1-A	P1-S	P1-S − P1-A
9×	18.3	22.7	**27.8**	+5.2
18×	10.8	13.0	**18.9**	+5.9
45×	7.8	10.3	**11.6**	+1.3
91×	4.6	7.4	**8.7**	+1.3

TABLE IV

ANSWER RECALL (%) WITH THE BUDGET DENOMINATED IN BYTES, CHARGING 1,536 B PER RETRIEVABLE ITEM FOR ITS EMBEDDING. P0 MAINTAINS NO INDEX AND IS THEREFORE FLAT.

Budget (KiB)	P0	RAG	P1-A	P1-S
32	**18.3**	5.0	5.9	15.1
64	18.3	11.0	9.4	**24.9**
128	18.3	20.0	11.9	**33.1**
256	18.3	33.0	15.8	**46.0**
512	18.3	50.3	22.2	**61.0**

TABLE V

COMPONENT ABLATION AT $S = 4096$, SHOWING THE CHANGE FROM THE FULL SYSTEM. †: THE ROW ON WHICH THE TWO METRICS DISAGREE IN DIRECTION.

Component removed	Δ evidence	Δ answer
Retrieval (Tier 3)	−31.6	−28.0
Demotion on pressure †	−11.5	+7.1
Scored eviction (→ FIFO)	−6.4	−7.4
Informative compressor	−4.4	−1.7
Consolidation merge	−3.4	−4.8
Semantic supersession	−0.3	−0.3

TABLE VI

SINGLE-COMPONENT SCORER SWAP AT $S = 4096$ WITH CONTEXT 2048. POLICY, RETRIEVAL AND COMPRESSION ARE FIXED AND ONLY THE SCORER VARIES. P3 IS FITTED ON THE OTHER NINE CONVERSATIONS. PRE-FIX RUN (SECTION VIII).

	What the scorer sees	Evid.	Ans.	Δ vs. P1	95% CI
P0	position only	12.6	25.6	−11.19	$[-17.24, -6.19]$
P1	regex + shallow NER	19.8	**36.8**	—	—
P2	the turn, in isolation	14.0	30.9	−5.87	$[-10.77, -1.76]$
P3	turn + fitted base rates	33.4	48.3	+11.47	$[+4.90, +17.86]$

evaluation [41], nonetheless loses: in the pre-fix run, Qwen2.5-0.5B-Instruct reaches 30.9% against the heuristic's 36.8%, -5.87 points with the interval excluding zero, at some fifteen minutes of GPU time per corpus against microseconds for a regular expression. And the headroom is real: a logistic scorer fitted under leave-one-conversation-out reaches 48.3%, $+11.47$ over the heuristic, so the judge was not limited by a ceiling that does not exist.

We attribute the judge's failure to what it is shown. It rates each turn in isolation, and salience is not a property of a turn considered alone: a phone number matters because a later turn asks for it. The fitted scorer sees the same isolated text but absorbs corpus-level base rates a per-turn prompt cannot convey — its strongest single feature is turn length, ahead of every hand-designed cue. Under leave-one-conversation-out it never sees the questions of the conversation it scores, so it is deployable in the way any supervised component is: fitted offline on labelled conversations, then applied to new ones. What this does not establish is that the base rates it absorbs carry over to a different kind of conversation, which Section VI-G tests.

G. A learned scorer across two corpora

LoCoMo's dialogues are generated by an LLM, so a scorer fitted to them may learn the generator's style rather than what people later ask about. We test this on REALTALK [34]: ten conversations between real people over 21 days, 8,944 turns and 660 questions with turn-level evidence in LoCoMo's schema, of which 160 are answer-recoverable. The scorer is a logistic regression over 15 shallow features of the message (length, digits, dates, capitalisation, question and filler cues) concatenated with its 384-dimensional MiniLM embedding, labelled by whether any question cites the turn as evidence. Every held-out conversation is scored by three models that never saw it: trained on the other conversations of the same corpus, on the other corpus only, or on all 19 other conversations pooled. Policy, budgets and retrieval are those of Section VI-B with the fixed heuristic (Section VIII); only the scorer varies.

Table VII reports the result. On LoCoMo the pooled scorer reaches 52.0% answer recall against the heuristic's 37.5%, a gain of $+14.55$ points, and pooling beats same-corpus training

TABLE VII

LEARNED SCORER ON HELD-OUT CONVERSATIONS OF TWO CORPORA, HAND FEATURES PLUS MINILM, $S = 4096$, CONTEXT 2048. POLICY P1-S; ONLY THE SCORER VARIES. Δ IS ANSWER RECALL AGAINST P1, WITH 95% INTERVALS RESAMPLED BY CONVERSATION; BOLD EXCLUDES ZERO.

Scorer (trained on)	AUC	Evid.	Ans.	Δ vs. P1	95% CI
LoCoMo (715 answer-recoverable questions)					
RAG (FIFO, no scorer)	–	17.8	28.1	—	—
P1 heuristic	0.599	20.2	37.5	—	—
P3, same corpus	0.793	32.9	48.7	**+11.19**	[+4.72, +17.06]
P3, other corpus only	0.692	30.1	46.6	**+9.09**	[+4.53, +13.46]
P3, pooled (shipped)	0.791	34.0	52.0	**+14.55**	[+10.40, +19.70]
REALTALK (160 answer-recoverable questions)					
RAG (FIFO, no scorer)	–	10.5	16.9	—	—
P1 heuristic	0.624	12.6	32.5	—	—
P3, same corpus	0.737	18.5	38.8	+6.25	[−3.01, +13.42]
P3, other corpus only	0.681	11.2	28.1	−4.38	[−12.50, 0.00]
P3, pooled (shipped)	0.749	17.3	36.2	+3.75	[−2.92, +8.65]

by $+3.36$ ($[+0.27, +6.52]$), so adding a different corpus helped rather than diluted. Transfer works in one direction only. Trained on REALTALK alone, the scorer still beats the heuristic on LoCoMo by $+9.09$; trained on LoCoMo alone, it falls below the heuristic on REALTALK (-4.38, n.s.) and is significantly worse than same-corpus training there (-10.62, $[-18.31, -3.45]$). At this budget no gain over the heuristic on REALTALK is significant, which with 160 questions in ten clusters is a statement about power as much as about the scorer.

$S = 4096$ is the budget of the headline comparison, not one chosen for this scorer, so Table VIII repeats the pooled run at four other budgets. On LoCoMo the gain is significant from $S = 2048$ to 16384 and largest at 4096; it shrinks at both ends, where the store holds too little for any ranking to matter or nearly everything anyway. On REALTALK the gain is significant at $S = 8192$ ($+16.88$, $[+6.31, +26.72]$) and 16384. The one budget at which neither corpus shows a significant gain is $S = 1024$, about 5% of a conversation.

The embedding is what carries across. With the hand features alone, the pooled scorer is significantly *worse* than the heuristic on REALTALK (-8.75, $[-17.45, -2.27]$): its feature weights encode LoCoMo's style, such as long turns ending in a period, which people messaging each other do not write. This is the same failure the heuristic shows on LongMemEval (Section IX). Logistic regression was kept after comparing it with boosted trees and a 64-unit MLP on the same folds: neither beat it by more than the spread between held-out conversations, and the trees were much slower both to fit and on the write path.

H. End-task accuracy

Every result above is context recall, which bounds end-task accuracy without measuring it. We therefore send each assembled context through one fixed reader, Qwen2.5-1.5B-Instruct-4bit decoding greedily for at most 32 tokens, and score the output with the same normaliser the answer-recall metric applies to the context. "The answer reached the context" and "the answer reached the output" are then one predicate evaluated at two points in the pipeline, so the ceiling comparison is exact rather than approximate. The run covers all 715 answer-recoverable questions with two controls: a closed-book arm that receives no memory at all, and an oracle arm that receives exactly the cited evidence.

TABLE VIII

THE POOLED LEARNED SCORER (P3) AGAINST THE P1 HEURISTIC ACROSS THE STORE BUDGET S, ANSWER RECALL (%), CONTEXT 2048, SAME PROTOCOL AS TABLE VII. Δ IS P3 MINUS P1 WITH 95% INTERVALS RESAMPLED BY CONVERSATION; BOLD EXCLUDES ZERO.

	LoCoMo			REALTALK		
S	P1	P3	Δ [95% CI]	P1	P3	Δ [95% CI]
1024	15.8	19.0	+3.22 [0.00, +6.40]	8.1	10.0	+1.88 [−1.95, +5.36]
2048	27.8	33.7	**+5.87** [+1.83, +9.88]	16.2	15.0	−1.25 [−10.62, +3.98]
4096	37.5	52.0	**+14.55** [+10.40, +19.70]	32.5	36.2	+3.75 [−2.92, +8.65]
8192	57.8	69.1	**+11.33** [+6.06, +16.41]	41.2	58.1	**+16.88** [+6.31, +26.72]
16384	71.3	77.2	**+5.87** [+2.04, +10.56]	61.3	68.1	**+6.88** [+2.12, +12.21]

TABLE IX

END-TASK ACCURACY THROUGH ONE FIXED READER AT $S = 4096$ WITH CONTEXT 2048, OVER ALL 715 ANSWER-RECOVERABLE QUESTIONS. ACC.|PRESENT IS ACCURACY RESTRICTED TO THE QUESTIONS WHOSE ANSWER REACHED THE CONTEXT; IT IS UNDEFINED FOR THE ARMS THAT NEVER PLACE ONE. PRE-FIX RUN (SECTION VIII).

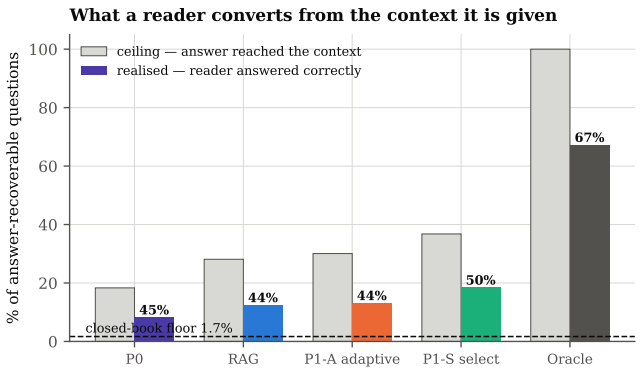
Arm	Acc.	F1	Ans. present	Acc.|present
Closed-book *(floor)*	1.7	5.0	0.0	–
P0 sliding window	8.3	11.9	18.3	40.5
RAG store-all	12.4	17.7	28.1	42.3
P1-A adaptive (compress)	13.1	19.9	30.1	42.3
P1-S select (drop)	**18.5**	24.7	36.8	49.4
Oracle *(ceiling)*	67.3	75.7	100.0	67.3

Table IX reports the outcome. The closed-book floor is 1.7%, so parametric knowledge contributes almost nothing on this benchmark and the margin above that floor is attributable to the memory layer. P1-S reaches 18.5% against RAG's 12.4%. Resampling by conversation as before, the realised gap is $+6.0$ points with a 95% interval of $[+3.3, +9.1]$, which is 69% of the $+8.7$-point recall gap between the same two arms in this pre-fix run (Section VIII).

Accuracy is not bounded by answer recall, since a reader may guess, so the table separates the two regimes. Accuracy given that the answer was present is 49.4% for P1-S against 42.3% for RAG, while accuracy given that it was absent is below 1% for every arm. The selection advantage therefore arrives through both available routes: P1-S places the answer in context more often, and the contexts it assembles are also the ones the reader uses more successfully. The oracle arm converts a complete context into a correct answer 67.3% of the time, which bounds what any memory policy can deliver through this reader and locates the remaining headroom in the reader rather than in the memory layer.

I. Against MemGPT-style and Mem0-style memory

Every baseline so far is internal. To place MemGate against designs in use, the memory management of MemGPT [8] and Mem0 [16] was rebuilt inside the same harness, at the same context budget and store budgets. The MemGPT-style



Percentages above the solid bars are the conversion rate: accuracy as a share of its own ceiling. Reader Qwen2.5-1.5B-Instruct-4bit, greedy, context 2048 / store 4096, scored on the 715 answer-recoverable questions.

Fig. 5. Reader accuracy by memory policy, with the closed-book floor and the oracle ceiling. The ordering matches the recall ordering, at a lower level.

arm keeps a recent queue and a searchable archive, and folds messages evicted from the archive into a recursive summary that is always in context and loses its oldest detail first. The Mem0-style arm writes one compressed fact per turn with ADD, UPDATE (the nearest stored fact above a similarity threshold is replaced, newest wins) and NOOP for filler, and evicts the least recently touched fact. Neither calls an LLM: summaries and facts are written by MemGate's own extractive compressor, so these are the systems' memory structures rather than their prompts. In the baselines' favour, each one's settings were grid-searched on LoCoMo itself at $S = 4096$ (MemGPT-style: summary share 0.25, 16-token gists; Mem0-style: 32-token facts, update threshold 0.65). No MemGate arm was tuned on these answers.

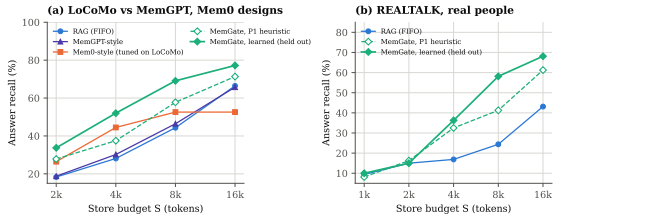
Table X and Fig. 6 report the result. The learned scorer beats both rebuilds at every budget, by $+7.3$ to $+24.6$ points over Mem0-style and $+11.5$ to $+22.7$ over MemGPT-style, every gap significant under clustering. The untrained P1 heuristic beats MemGPT-style everywhere ($+5.6$ to $+11.3$) but loses to the tuned Mem0-style at $S = 4096$, the budget its settings were chosen on, by 7.0 points with the interval excluding zero; it is level at 2048, ahead at 8192 by $+5.2$ (McNemar $p = 0.03$, but not significant once resampled by conversation), and well ahead at 16384. Mem0-style stops improving at 52.6% from $S = 8192$: UPDATE replaces a fact with its nearest neighbour, so a threshold loose enough to help at small S discards distinct facts that merely look alike, and the store never fills. MemGPT-style stays within about two points of plain RAG, because its summary holds gists whose answers were compressed away, the failure of Section VI-B in a different form.

J. Chats of up to ten million tokens

LoCoMo conversations average about 19K tokens. BEAM [47] supplies synthetic user–assistant chats at 100K, 500K, 1M and 10M tokens with probing questions that cite turns, and MemBench [48] short agent trajectories made of factual statements. Two repairs were needed before BEAM 1M and 10M could be scored: some chats restart their turn numbering

TABLE X
ANSWER RECALL (%) ON LOCOMO AGAINST REBUILT MEMGPT-STYLE AND MEM0-STYLE MEMORY, CONTEXT 2048. THE BASELINES' SETTINGS WERE TUNED ON LOCOMO AT $S = 4096$; P1-S AND THE HELD-OUT LEARNED SCORER P3 WERE NOT. BOLD Δ: 95% INTERVAL RESAMPLED BY CONVERSATION EXCLUDES ZERO.

		MemGPT	Mem0			Δ vs. Mem0-style	
S	RAG	-style	-style	P1-S	P3	P1-S	P3
2048	18.3	18.7	26.4	27.8	**33.7**	$+1.4$	$\mathbf{+7.3}$
4096	28.1	30.2	44.5	37.5	**52.0**	$\mathbf{-7.0}$	$\mathbf{+7.6}$
8192	44.3	46.4	52.6	57.8	**69.1**	$+5.2$	$\mathbf{+16.5}$
16384	66.4	65.7	52.6	71.3	**77.2**	$\mathbf{+18.7}$	$\mathbf{+24.6}$



Context budget 2048. The learned scorer is fitted per held-out conversation (the other conversations plus the other corpus). The baselines' knobs were grid-searched on LoCoMo at S=4k, which favours them. LLM-free rebuilds of each system's memory management (memgate/external_baselines.py), not the products.

Fig. 6. Answer recall against the store budget for MemGate and the rebuilt MemGPT-style and Mem0-style memories (Table X).

partway, so each question is pinned to the segment whose cited turns share the most words with it, and only when that segment leads clearly. At 1M, 3 of 625 questions could not be pinned; at 10M, 128 of 200 are pinned and the rest are abstentions or cite turns that no part of the chat holds. Each conversation is then scored by a learned scorer fitted, under five folds by conversation, without it; BEAM 1M and 10M are outside its training corpora altogether. The store budget is $S = 26{,}000$ for BEAM, about 2.5% of a 1M-token chat.

Table XI reports answer recall. The learned scorer beats RAG on all seven corpora with the interval excluding zero, by $+36.5$ points on BEAM 1M and $+46.2$ on BEAM 10M. On a million-token chat RAG keeps only the most recent turns, so almost nothing it is asked about survives: evidence recall, which covers 622 questions on BEAM 1M rather than 52, is 22.2% for the learned scorer against 0.9% for RAG. The one corpus on which it is not the best is MemBench, whose turns are short factual statements, which is what the P1 rules were written to find; there P1 leads it by 1.9 answer points (n.s.). BEAM answers are mostly paraphrases, so only a few questions per corpus are answer-recoverable, and the BEAM 10M row rests on 13 questions.

K. Ten times the training data

The learned scorer that the application first shipped was fitted on 58,616 turns. We retrained it on ten times that: BEAM 1M (74,630 turns) and 10M (208,696), and MemBench's observation and participation sets, 594,000 turns in all, with the families of Eq. (4) weighted equally. Each candidate was replayed through P1-S on held-out conversations at three to five budgets per corpus, and the whole protocol was repeated

TABLE XI

HELD-OUT ANSWER RECALL (%) AT EACH CORPUS'S OWN STORE BUDGET, 5-FOLD BY CONVERSATION. $P3_{12}$ IS THE PR #12 RECIPE (LOCOMO, REALTALK, BEAM 100K/500K), REFITTED WITHOUT THE HELD-OUT FOLD; $P3_{ship}$ IS THE SHIPPED ENSEMBLE OF SECTION VI-K, UNDER THE SAME PROTOCOL. n IS THE NUMBER OF ANSWER-RECOVERABLE QUESTIONS. BOLD: THE 95% INTERVAL, RESAMPLED BY CONVERSATION, EXCLUDES ZERO. ‡PERSONACHAT VALIDATION DIALOGUES, NEVER TRAINED ON; QUERIES ARE THE PERSONA LINES, A PROXY DESCRIBED IN SECTION VI-K.

Corpus	S	n	RAG	P1	$P3_{12}$	$P3_{12}-$RAG [95% CI]	$P3_{ship}$	$P3_{ship}-P3_{12}$ [95% CI]
LoCoMo	4096	715	28.1	37.5	50.3	**+22.2** [+17.5, +27.6]	51.9	+1.5 [−0.2, +3.4]
REALTALK	4096	160	16.9	32.5	36.2	**+19.4** [+7.5, +26.4]	36.2	0.0 [−3.3, +3.1]
BEAM 100K	26K	34	32.4	35.3	73.5	**+41.2** [+14.3, +67.6]	73.5	0.0 [0.0, 0.0]
BEAM 500K	26K	55	32.7	32.7	63.6	**+30.9** [+16.7, +44.8]	61.8	−1.8 [−10.0, +5.8]
BEAM 1M	26K	52	13.5	11.5	50.0	**+36.5** [+21.8, +51.0]	50.0	0.0 [−7.3, +8.0]
BEAM 10M	26K	13	7.7	23.1	53.8	**+46.2** [+10.0, +76.9]	38.5	−15.4 [−44.4, 0.0]
MemBench	2048	367	76.8	97.5	95.6	**+18.8** [+14.2, +23.7]	97.5	**+1.9** [+0.3, +3.8]
Everyday chat‡	1024	1762	28.8	30.4	61.4	—	69.0	**+7.6** [+5.9, +9.5]

with the folds dealt from three different orders, 26,444 replays in all. The rule for shipping a model is that it must lose nowhere.

More data alone did not help (Table XII, first column). The 10x model gains on MemBench, the corpus that was added, but loses evidence recall on BEAM 1M by 5.1 to 5.6 points and on BEAM 500K by 3.0 to 5.2, with the interval excluding zero in every shuffle. Our reading is that a linear model on 399 inputs was not short of data but of capacity: the new turns are a different kind of chat, mostly negatives (BEAM 10M adds 208,696 turns and only a few hundred evidence labels), and they pull the weights toward short factual statements at the cost of BEAM's long answers. Boosted trees, the obvious source of capacity, gained on BEAM 500K but lost 8.5 points on LoCoMo ($[-11.1, -5.3]$) and were rejected.

Three changes fixed it. The family weight of Eq. (4) is split among corpora by evidence turns instead of equally, so BEAM 10M's many unlabelled turns count for little. Everyday chat is added as a fifth family: 894 long chats assembled from PersonaChat [49], 262,876 turns, weakly labelled (a turn is evidence when it states one of its speaker's persona lines). And the result is averaged in logit space with the earlier model, Eq. (5), which keeps the ranking that suited BEAM's long answers while adding what the new chats taught, at a linear model's cost. The shipped model is trained on 856,876 turns, 14.6 times the original. For an everyday-chat test, 50 PersonaChat validation chats that no model saw are replayed at $S = 1024$; each stated persona line is asked as a query, and the answer is the words of it that the person said. That is a proxy, since query and evidence share wording, but it treats every arm alike.

At the paper's budgets the shipped model has no significant loss on any corpus in any of the three shuffles (Table XII). It gains on everyday chat (+7.6 to +8.2 answer points, 69.0% against 61.4%), on MemBench, and on BEAM 100K through the application's own path (+11.5 to +12.4 evidence points), and ties on BEAM 1M, where the 10x model lost. Across all other budgets it has one significant loss: BEAM 1M evidence recall at $S = 8000$, -1.7 $[-2.8, -0.6]$, about one cited turn in sixty.

TABLE XII

RETRAINING: CHANGE AGAINST THE PR #12 RECIPE IN THREE RESHUFFLED FOLD ORDERS, AT EACH CORPUS'S BUDGET (TABLE XI), HELD OUT. "APP PATH" ADDS THE APPLICATION'S CHUNKING, GUARD AND MULTI-QUERY RETRIEVAL. BOLD: THE 95% INTERVAL, RESAMPLED BY CONVERSATION, EXCLUDES ZERO.

	10x (594K turns)	Shipped (857K turns)	
Corpus	Evidence	Answer	Evidence
LoCoMo	−0.4, −0.8, −0.2	+1.5, −0.3, +0.3	+0.1, −0.2, +0.2
REALTALK	+0.1, −1.6, 0.0	0.0, 0.0, +0.6	−0.3, −0.8, +0.2
BEAM 100K	−1.3, −0.8, **−3.7**	0.0, +2.9, −2.9	+1.8, **+1.8**, +0.6
BEAM 500K	**−5.2**, **−4.4**, **−3.0**	−1.8, +1.8, +1.8	+1.6, +2.0, **+2.8**
BEAM 1M	**−5.1**, **−5.6**, **−5.3**	0.0, −1.9, +1.9	+0.3, 0.0, −0.4
BEAM 10M	**−2.3**, −2.5, −0.5	−15.4, 0.0, −7.7	−1.2, −1.8, +1.2
MemBench	**+5.5**, **+5.8**, **+6.2**	**+1.9**, **+1.9**, **+2.2**	**+5.6**, **+5.9**, **+6.4**
Everyday chat‡	—	**+7.6**, **+8.2**, **+8.0**	**+8.8**, **+9.1**, **+9.3**
App path, BEAM 100K	**+6.1**, +4.2, +6.0	**+11.8**, **+11.8**, +8.8	**+12.4**, **+11.5**, **+11.9**
App path, LoCoMo	−0.5, −0.7, −0.2	+1.5, 0.0, +0.4	+0.3, +0.1, +0.5
App path, REALTALK	+0.7, +0.3, −0.3	+3.1, +1.9, +0.6	+0.5, −0.1, 0.0

VII. FROM HARNESS TO APPLICATION

The policy is also a working system. A chat application wraps a reader model in the same `observe()` and `build_context()` calls the experiments use (Fig. 1): every message, the user's and the model's reply alike, passes through the policy, and each question is answered from the context MemGate assembles, never from the raw history. The application adds one step the experiments do not have, an entry gate that scores each message the moment it is said. A message the scorer labels filler (a greeting, "ok cool", at most two words), or a model reply scoring below 0.1, is forgotten at once and never takes space. Under P1-A and P1-S a message with $u \geq \tau_{\text{fact}}$ is also written to Tier 3 at once, with its score, while it stays in Tier 1 for the next few replies; when it later leaves Tier 1, the near-duplicate rule above folds it into that copy. Every other message follows the experimental path unchanged. The gate can be switched off, and the built-in benchmark below runs the policies without it. The reader is interchangeable because it only ever receives two strings, the assembled context and the current message. The default runs offline on an 8 GB laptop with Qwen2.5-1.5B-Instruct [42]; a hosted model (Claude, or Gemini [50] on its free tier) can be substituted, and when a hosted model is rate-limited and a local copy is installed, Qwen answers that message instead. With no model at all the application shows the context

TABLE XIII

WHOLE BEAM CHATS TYPED INTO THE APPLICATION ONE MESSAGE AT A TIME, $S = 26{,}000$. N CHATS OF THE GIVEN MEAN LENGTH IN MILLIONS OF TOKENS; n ANSWERABLE QUESTIONS; ANSWER RECALL (%). BEAM 1M IS OUTSIDE THE SCORER'S TRAINING CORPORA; BEAM 500K IS INSIDE THEM.

Chats	N	M tok.	n	P0	RAG	MemGate	Δ vs. RAG [95% CI]	McNemar p
BEAM 1M	33	1.07	52	9.6	19.2	**51.9**	**+32.7** [+19.2, +46.3]	9.1×10^{-4}
BEAM 500K	33	0.52	55	12.7	30.9	**65.5**	**+34.5** [+19.6, +49.1]	3.1×10^{-4}

the reader would have received. The application runs P1-S by default, with the shipped learned scorer of Section VI-K inside GuardedScorer (Eqs. (5)–(7), Fig. 2); if the MiniLM embedder is unavailable it falls back to the P1 heuristic rather than scoring with the wrong features. Any text can be added: pasted text, PDF, Word, PowerPoint and other files are cleaned, and anything over 120 tokens is cut into pieces of at most 48 tokens, each scored, stored and retrieved like a turn, because a single embedding of a long document describes only its opening.

Two properties follow from reusing the experimental code. First, persistence is exact: a conversation is saved as its messages and restored by replaying them through `observe()`, which is deterministic, so the tiers after a restart are the tiers before it. Second, the application can check itself: a built-in benchmark run recomputes the RAG and P1-S arms on all of LoCoMo at $S = 4096$ on a CPU and compares them with the committed results, reproducing 28.1% and 37.5% exactly. A side panel shows each message's utility u, the tier it was routed to or the reason it was forgotten, and the exact context sent to the reader, so every decision the paper measures in aggregate can be inspected one message at a time. The application runs at the experimental budgets by default and offers a reduced scale (Tier 1 of four messages, a 512-token context and $S = 150$) under which eviction becomes visible within a few messages; these sizes and the entry gate are the only settings that differ from the experiments.

A. Stress and scale through the application

The benchmarks above replay conversations through the policy. To test the system a person would use, whole BEAM chats were typed into the application one message at a time, exactly as its page sends them, with each chat's recorded assistant turns as the replies, at $S = 26{,}000$. Each answerable question was then asked as a message, and the notes each memory would hand the reader were checked. Table XIII reports the result: on million-token chats, which MemGate holds at about 2.5%, the answer reaches the reader for 51.9% of questions against 19.2% for RAG and 9.6% for a sliding window. The store never exceeded S, and a message took 0.49 s on average on a shared 4-core machine. This run used the PR #12 scorer, the one shipped at the time; repeating it with the current one takes about ten hours and was not done, but the held-out application-path rows of Table XII are the current model's.

Where the evidence sits explains much of the gap. Table XIV splits LoCoMo and REALTALK questions by the position of their evidence in the conversation. RAG forgets oldest-first, so for evidence in the first third it keeps the answer 8–9% of the time; the untrained P1 heuristic keeps 20–33%, and the learned scorer more, although on these two corpora it is in-sample and its column is a ceiling. For evidence in the last third RAG is competitive, since recency is then the right rule.

TABLE XIV

ANSWER RECALL (%) BY THE POSITION OF A QUESTION'S EVIDENCE IN THE CONVERSATION, $S = 4096$. RAG AND P1 ARE UNTRAINED. *THE SHIPPED PR #12 SCORER, IN-SAMPLE ON BOTH CORPORA.

Corpus	Evidence in	n	RAG	P1	P3*
LoCoMo	first	262	9.2	33.2	51.9
	middle	225	14.2	38.2	53.8
	last	228	63.6	41.7	54.4
REALTALK	first	86	8.1	19.8	33.7
	middle	45	11.1	40.0	55.6
	last	29	51.7	58.6	65.5

People do not ask one clean question per message. When each LoCoMo question was buried in 150–450 words of other chat, using the whole message as the query dropped answer recall from 53.1% to 30.3%; joining the message's questions into one query recovered 42.1%, and making each question its own query with interleaved rankings (Eq. (11)) recovers 49.8% (REALTALK: 47.5%, 26.2%, 42.5% and 43.8%).

The same tests found six input-handling bugs, each now pinned by a test: half of an emoji in one message made every later save of that chat fail; a 300 KB Word file that unzips to 300 MB used 3.8 GB of memory; deeply nested JSON dropped the connection; request bodies had no size limit; a message of only invisible characters was accepted; and a locked PDF gave an unclear error. A load test found one more: under bursts, the server's accept queue of five reset 314 of 2,000 requests from 200 simultaneous visitors. With a queue of 256 the same test completes with no errors in 25 s (95th percentile 2.7 s per message), and 100 messages sent into one chat at the same moment are all kept. A single chat of 5,000 messages stays at 4,093 of its 4,096 tokens, at 0.12 s per message by the end. The one slow path left is reopening a saved million-token chat, which replays it through memory and takes 70 s.

VIII. THREATS TO VALIDITY

Three errors changed our results before we detected them. Each was caught by an invariant rather than by reading the code. Two are forms of leakage, a known cause of irreproducible machine-learning results [51], [52].

A policy that read its evaluation label. LoCoMo supplies a `fact_id` per evidence turn, and an early consolidation rule promoted items using it — the policy was consulting the answer key. The effect was worth 45 to 47 recall points and invalidated our first reported result. The remedy is an invariant, not a convention: strip every evaluation label from the input and the resulting store must be bit-identical. Each decision

point added since was placed under it before any number was reported.

Component value depends on the current bottleneck. We first measured the sentence embedder at $+0.2$ points and concluded it barely mattered. That held only in the regime measured, where retrieval was saturated against a write path discarding 93% of the conversation; once that bottleneck was removed the same substitution was worth $+19.7$. Ablations run against a bottlenecked pipeline systematically undervalue whatever would matter after the dominant term moves.

A failure invisible in the aggregate. Batched judge scoring returned `NaN` for short turns specifically. Precision was the natural hypothesis and `float32` did not help; the cause was left-padding, producing fully masked attention rows. Without an explicit guard every filler turn would have scored `NaN` and inverted the eviction order.

A fourth defect was found after the main runs: the heuristic scored each turn with its speaker tag, which blocked the filler rule and gave nearly every turn the proper-noun bonus (5,863 of 5,882 LoCoMo turns were labelled facts). Recall tables and Table VII use the fixed scorer; the judge, end-task, abstractive and LongMemEval runs predate it. The fix moves the headline P1-S result from 36.8% to 37.5%, and it shrinks the learned scorer's lead over the heuristic at $S = 2048$ from $+5.2$ to $+2.7$ points, because part of that lead was the defect handicapping the heuristic.

IX. Limitations

Few conversations. Clustering by conversation is the correct unit and leaves few clusters, so intervals are wide; we report the effects that do not survive as not significant. REALTALK adds ten conversations but only 160 answer-recoverable questions, too few to resolve most of its differences.

A single reader model. Section VI-H uses one 1.5B four-bit model [42] decoding greedily. We claim the ordering of policies, not the absolute accuracies: the oracle arm's 67.3% shows much of the remaining error is the reader, and a larger one would raise every arm and could compress or widen the margins; a larger reader may also use position in the context differently [5].

Compression was tested in both forms. Replacing P1-A's word-selection with a 1.5B model rewriting each evicted turn is worth $+4.06$ answer points (CI $[+0.84, +6.66]$), so part of the negative result was the compressor rather than compression. Selection still wins (pre-fix run): 36.8% against 34.1%, at 2.1 s against 3,892 s and 4,879 generations. The claim narrows to: a better compressor helps, and does not overtake dropping.

Concatenated streams are synthetic, as discussed in Section VI-D.

The scorer is tuned to this corpus. On a 94-item LongMemEval [6] subsample, whose turns average 210 tokens against LoCoMo's 32, the heuristic's separation between evidence and non-evidence turns reversed in sign and scored eviction lost to FIFO. Read the $+9.37$ of Section VI-B as what a policy recovers *given a scorer suited to the corpus*, not as a property of scored selection in general. The learned scorer has since been tested held out on BEAM, where the heuristic also ranks evidence below chance, and on MemBench and everyday chat (Section VI-J); LongMemEval, where the heuristic's sign first reversed, is still untested with it, because it could not be fetched in our environment. Its labels are distant supervision (a turn is positive if any question cites it), and that supervision is twice as dense on LoCoMo (24% of turns) as on REALTALK (12%); on PersonaChat it is weaker still, a word-overlap match with a persona line.

Few answerable questions at scale. BEAM's answers are mostly paraphrases, so only 34 to 55 questions per scale are answer-recoverable, and 13 at 10M. The intervals of Table XI are correspondingly wide, and evidence recall, which covers about ten times as many questions, is reported alongside. The rebuilt MemGPT-style and Mem0-style arms reproduce those systems' memory structures without their LLM calls, so they measure the designs, not the products.

Single embedder and single judge. The negative result of Section VI-F is evidence about a 0.5B model prompted per turn, not about LLM-judged salience in general. A larger judge, or one shown the neighbouring turns, remains untested and is the most direct way to challenge the finding.

X. Conclusion

Under the accounting normally used here, tiered compression is a net loss against storing everything and retrieving — but that is an artefact of leaving the store unbounded. Bound it, and the decision of which items to forget is worth $+9.37$ answer-recall points at identical storage, fidelity and retrieval, while compressing an evicted item rather than dropping it loses 7.13. The crossover at which rate–distortion theory [12] says compression should begin to pay is not reached to $182\times$. Charging the budget for the embedding index reverses the ranking below about 40 KiB. An LLM judge, the obvious route to a better scorer, loses to a regular expression, for an identifiable reason: salience is not a property of a turn in isolation. A logistic scorer over a sentence embedding, trained on LLM-generated and human conversations together, beats the heuristic by $+14.55$ points on held-out LoCoMo conversations, and by a significant margin at every store budget from 2048 to 16384 tokens; it transfers from human dialogue to generated dialogue but not the other way. Through a fixed reader, $+6.0$ of the pre-fix $+8.7$ points survive as correct answers. The same scorer beats MemGPT-style and Mem0-style memory, rebuilt in this harness and tuned in their favour, at every budget tested, and on held-out chats of up to ten million tokens it beats retrieval on every corpus; typed into the application, million-token chats keep the answer 51.9% of the time against 19.2%. Ten times the training data did not by itself make a better linear scorer; weighting corpora by their labels, adding everyday chat and averaging with the earlier model did, without a significant loss at the paper's budgets.

The practical recommendation is narrow. Retain a scored subset at full fidelity rather than compressing everything [18], [21], and denominate the budget in bytes with the index

included. The methodological recommendation is broader. Three of the results above changed in sign or in magnitude once a measurement error was removed, and none of those errors was visible in the aggregate numbers they affected.

Two questions remain. The pooled scorer's gains on human conversation are significant only once the store holds about 40% of a conversation, and it is untested on LongMemEval, where hand-crafted cues inverted; more human-written evidence, rather than a larger model, is what would settle both. And a scorer learned online from retrieval feedback is already statistically level with the heuristic on LoCoMo without labels or future questions (-2.94, n.s.); whether it can be seeded with the pooled scorer and then adapt to each user is the step from a scorer fitted to benchmarks to one fitted to the person using the application.

Acknowledgment

The authors thank the faculty of SVKM's NMIMS, Indore Campus, for their guidance throughout this capstone project.

References

[1] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023.

[2] L. Wang, C. Ma, X. Feng, Z. Zhang, H. Yang, J. Zhang, Z. Chen, J. Tang, X. Chen, Y. Lin, W. X. Zhao, Z. Wei, and J.-R. Wen, "A survey on large language model based autonomous agents," *Frontiers of Computer Science*, vol. 18, no. 6, p. 186345, 2024.

[3] T. B. Brown, B. Mann, N. Ryder *et al.*, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.

[4] Chroma Research, "Context rot: How increasing input tokens impacts LLM performance," Chroma, Tech. Rep., 2025.

[5] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[6] D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu, "LongMemEval: Benchmarking chat assistants on long-term interactive memory," *arXiv preprint arXiv:2410.10813*, 2024.

[7] T. R. Sumers, S. Yao, K. Narasimhan, and T. L. Griffiths, "Cognitive architectures for language agents," *Transactions on Machine Learning Research*, 2024.

[8] C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez, "MemGPT: Towards LLMs as operating systems," *arXiv preprint arXiv:2310.08560*, 2023.

[9] W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang, "MemoryBank: Enhancing large language models with long-term memory," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 38, no. 17, 2024, pp. 19 724–19 731.

[10] Z. Zhang, X. Bo, C. Ma, R. Li, X. Chen, Q. Dai, J. Zhu, Z. Dong, and J.-R. Wen, "A survey on the memory mechanism of large language model based agents," *arXiv preprint arXiv:2404.13501*, 2024.

[11] C. E. Shannon, "Coding theorems for a discrete source with a fidelity criterion," in *IRE National Convention Record*, vol. 4, 1959, pp. 142–163.

[12] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley, 2006.

[13] R. Colaco and A. Lahjouji, "A rate–distortion framework for memory in language agents," *arXiv preprint arXiv:2607.08032*, 2026.

[14] J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proc. ACM Symp. User Interface Software and Technology (UIST)*, 2023.

[15] W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang, "A-MEM: Agentic memory for LLM agents," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[16] P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav, "Mem0: Building production-ready AI agents with scalable long-term memory," *arXiv preprint arXiv:2504.19413*, 2025.

[17] R. C. Atkinson and R. M. Shiffrin, "Human memory: A proposed system and its control processes," *Psychology of Learning and Motivation*, vol. 2, pp. 89–195, 1968.

[18] H. Jiang, Q. Wu, C.-Y. Lin, Y. Yang, and L. Qiu, "LLMLingua: Compressing prompts for accelerated inference of large language models," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 13 358–13 376.

[19] J. Mu, X. L. Li, and N. Goodman, "Learning to compress prompts with gist tokens," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[20] A. Chevalier, A. Wettig, A. Ajith, and D. Chen, "Adapting language models to compress contexts," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[21] T. Ge, J. Hu, L. Wang, X. Wang, S.-Q. Chen, and F. Wei, "In-context autoencoder for context compression in a large language model," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.

[22] G. Xiao, Y. Tian, B. Chen, S. Han, and M. Lewis, "Efficient streaming language models with attention sinks," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2024.

[23] Z. Zhang, Y. Sheng, T. Zhou, T. Chen, L. Zheng, R. Cai, Z. Song, Y. Tian, C. Ré, C. Barrett, Z. Wang, and B. Chen, "H2O: Heavy-hitter oracle for efficient generative inference of large language models," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[24] J. W. Rae, A. Potapenko, S. M. Jayakumar, C. Hillier, and T. P. Lillicrap, "Compressive transformers for long-range sequence modelling," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020.

[25] Y. Wu, M. N. Rabe, D. Hutchins, and C. Szegedy, "Memorizing transformers," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[26] W. Wang, L. Dong, H. Cheng, X. Liu, X. Yan, J. Gao, and F. Wei, "Augmenting language models with long-term memory," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[27] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[28] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[29] S. Borgeaud, A. Mensch, J. Hoffmann *et al.*, "Improving language models by retrieving from trillions of tokens," in *Proc. Int. Conf. Machine Learning (ICML)*, ser. PMLR, vol. 162, 2022, pp. 2206–2240.

[30] L. A. Belady, "A study of replacement algorithms for a virtual-storage computer," *IBM Systems Journal*, vol. 5, no. 2, pp. 78–101, 1966.

[31] P. J. Denning, "The working set model for program behavior," *Communications of the ACM*, vol. 11, no. 5, pp. 323–333, 1968.

[32] A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang, "Evaluating very long-term conversational memory of LLM agents," in *Proc. Annu. Meeting Assoc. Computational Linguistics (ACL)*, 2024.

[33] Y. Bai, X. Lv, J. Zhang *et al.*, "LongBench: A bilingual, multitask benchmark for long context understanding," in *Proc. Annu. Meeting Association for Computational Linguistics (ACL)*, 2024, pp. 3119–3137.

[34] D.-H. Lee, A. Maharana, J. Pujara, X. Ren, and F. Barbieri, "REALTALK: A 21-day real-world dataset for long-term conversation," *arXiv preprint arXiv:2502.13270*, 2025.

[35] W. Wang, F. Wei, L. Dong, H. Bao, N. Yang, and M. Zhou, "MiniLM: Deep self-attention distillation for task-agnostic compression of pre-trained transformers," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 5776–5788.

[36] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2019.

[37] M. Mintz, S. Bills, R. Snow, and D. Jurafsky, "Distant supervision for relation extraction without labeled data," in *Proc. Joint Conf. 47th Annu. Meeting ACL and 4th IJCNLP*, 2009, pp. 1003–1011.

[38] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[39] J. C. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*. MIT Press, 1999, pp. 61–74.
[40] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, "SQuAD: 100,000+ questions for machine comprehension of text," in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2016, pp. 2383–2392.
[41] L. Zheng, W.-L. Chiang, Y. Sheng *et al.*, "Judging LLM-as-a-judge with MT-Bench and Chatbot Arena," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.
[42] Qwen Team, "Qwen2.5 technical report," *arXiv preprint arXiv:2412.15115*, 2025.
[43] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall, 1993.
[44] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.
[45] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.
[46] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Bootstrap-based improvements for inference with clustered errors," *Review of Economics and Statistics*, vol. 90, no. 3, pp. 414–427, 2008.
[47] M. Tavakoli *et al.*, "Beyond a million tokens: Benchmarking and enhancing long-term memory in LLMs," in *International Conference on Learning Representations (ICLR)*, 2026, arXiv:2510.27246.
[48] H. Tan *et al.*, "MemBench: Towards more comprehensive evaluation on the memory of LLM-based agents," in *Findings of the Association for Computational Linguistics: ACL 2025*, 2025, arXiv:2506.21605.
[49] S. Zhang, E. Dinan, J. Urbanek, A. Szlam, D. Kiela, and J. Weston, "Personalizing dialogue agents: I have a dog, do you have pets too?" in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018, pp. 2204–2213.
[50] Gemini Team, Google, "Gemini: A family of highly capable multimodal models," *arXiv preprint arXiv:2312.11805*, 2023.
[51] S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, "Leakage in data mining: Formulation, detection, and avoidance," *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 4, pp. 1–21, 2012.
[52] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," *Patterns*, vol. 4, no. 9, p. 100804, 2023.